



Learning in Immersive Museum Environments

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Abstract

Due to the COVID-19 pandemic, museum professionals have adopted various technological resources that have expanded museums into new virtual spaces. These virtual spaces do much more than simply communicate information to visitors and attract them to visit the museum physically: they offer new teaching and learning contexts. The emergence of these new learning contexts calls for the development of new methodologies to analyse them. This presentation will focus on a study, published in the *International Journal of Art & Design Education*, whose goal was to develop a coding rubric for the systematic analysis of complex, interactive, immersive museum environments (Huebner, 2025). To achieve this, I adapted and tested Gillian Rose's (2022, *Visual methodologies: an introduction to researching with visual materials*) critical analysis approach to visual content to explore the museum educational possibilities of virtual reality, using immersive environments by the Louvre and the Victoria and Albert Museum as case studies. I developed a comprehensive coding grid, adjusting Rose's (2022) principles and modalities to the unique characteristics of virtual reality experiences as identified by Fegely and Cherner (2021, *Journal of Information Technology Education: Research*) and Lee and Cherner (2015, *Journal of Information Technology Education: Research*). The findings reveal numerous opportunities to leverage virtual reality's distinct features, depending on the subject matter, the virtual teaching environment, and the pedagogical strategies employed. The research and analytical grid promise to make a valuable contribution to the fields of museum education and museology because they provide a structured means to scrutinise and evaluate the educational possibilities inherent in virtual reality experiences while also deepening our understanding of the intricate dynamics between users and museum artworks as well as objects within immersive environments.



Fig. 1 Screenshot of Curious Alice, a VR experience created by the V&A and HTC Vive Arts. Featuring original artwork by Kristjana S Williams, 2020. © Victoria and Albert Museum, London.

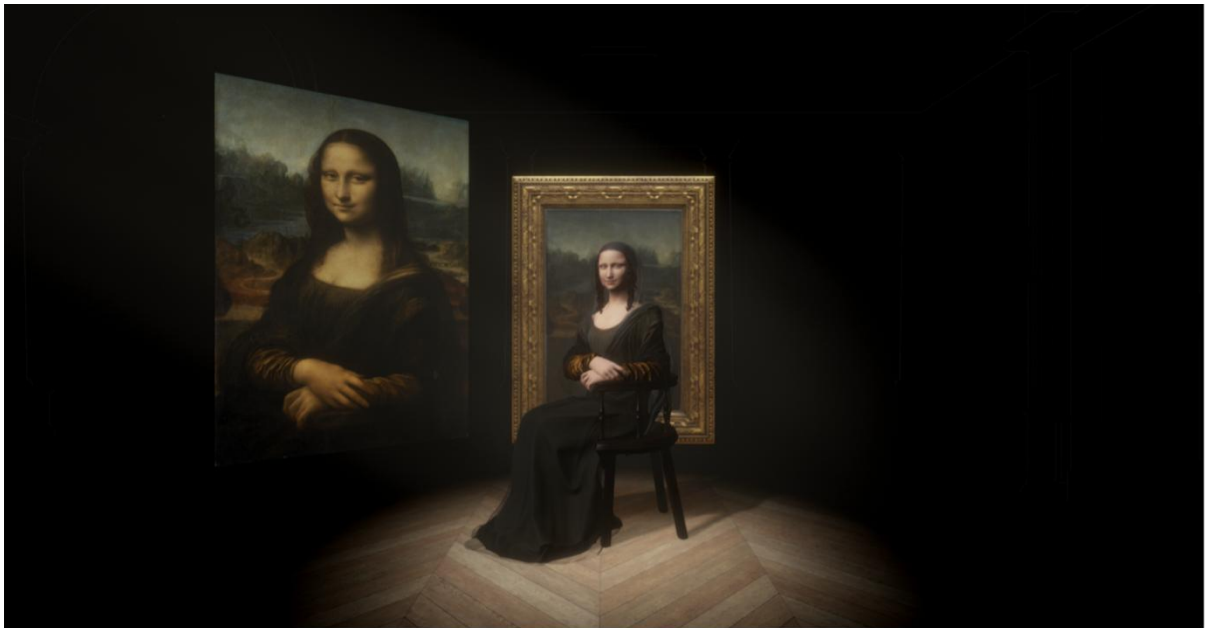


Fig. 2 Screenshot of Mona Lisa: Beyond the Glass, a VR experience created by the Louvre and Emissive featuring Leonardo da Vinci's artwork. © HTC – Musée du Louvre.

References

1. Fegely, A., & S Cherner, T. (2021). A Comprehensive Rubric for Evaluating EduVR. *Journal of Information Technology Education: Research*, 20, 137–171.
<https://doi.org/10.28945/4737>
2. Huebner, E.J. (2026), A Critical Analysis of Immersive Environments: A Methodology for Museum Education. *The International Journal of Art and Design Education*, 45, 60-76. <https://doi.org/10.1111/jade.12613>
3. Lee, C.- Y., & Cherner, T. S. (2015). A Comprehensive Evaluation Rubric for Assessing Instructional Apps. *Journal of Information Technology Education: Research*, 14, 021–053.
4. Rose, G. (2022). *Visual methodologies: An introduction to researching with visual materials* (Fifth). Sage Publications.